VICTORY 2027

TRANSPIRATION

LECTURE-2 BIOLOGY

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TOPICS TO BE COVERED

- Stomatal Transpiration
- [redacted]
- [redacted]

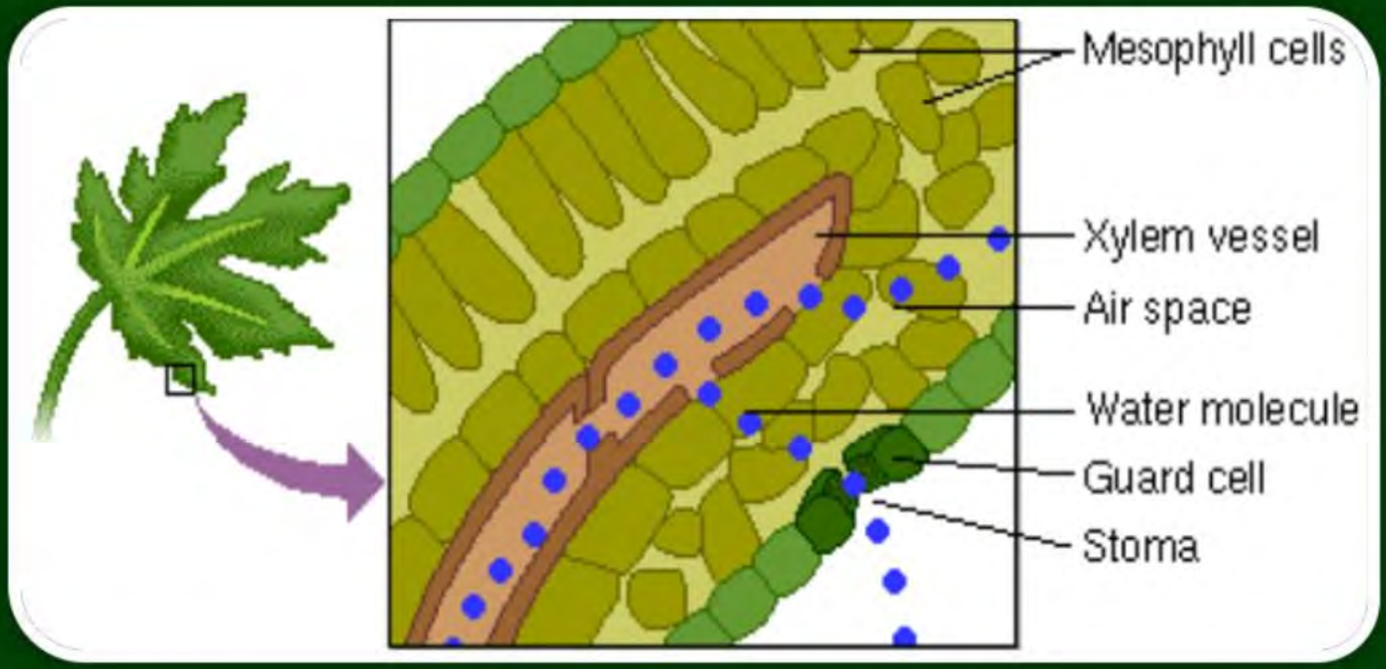
LENTICULAR TRANSPIRATION

- Lenticels are special openings that develop on the barks of older stems.
- These allow diffusion of gases for respiration.
- Lenticels never close. They remain open all the time.
- Water from the cell surfaces directly facing the lenticel evaporates and contributes to transpiration

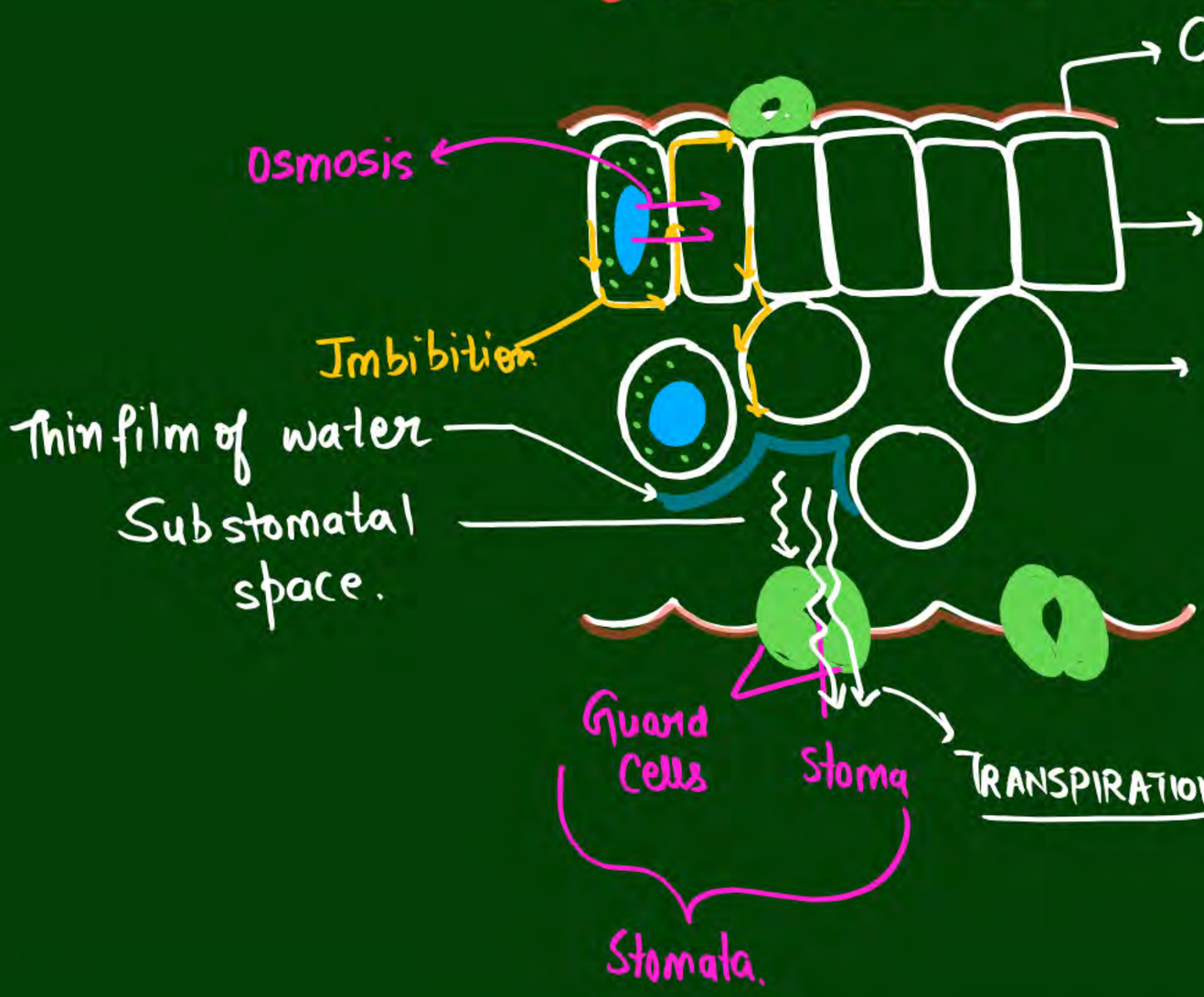


MECHANISM OF STOMATAL TRANSPIRATION

In stomatal transpiration, the water vapour escapes through the stomata of the leaf. Water, after absorption by the roots from the soil, rises up through the stem and reaches the tissues of the leaves through veins.



DICOT LEAF



Adaxial Surface [less Stomata]

TRANSPIRATION

Abaxial Surface [More number of stomata]

Steps of Stomatal Transpiration :-

1) All palisade and spongy cell will become turgid once the water is absorbed by OSMOSIS & IMBIBITION.

2) Because of turgor pressure, cells will lose water and form a thin film of water.

3) Thin film of water is formed at sub-stomatal space.

4) The guard cells become turgid to open stomata and take in CO_2.

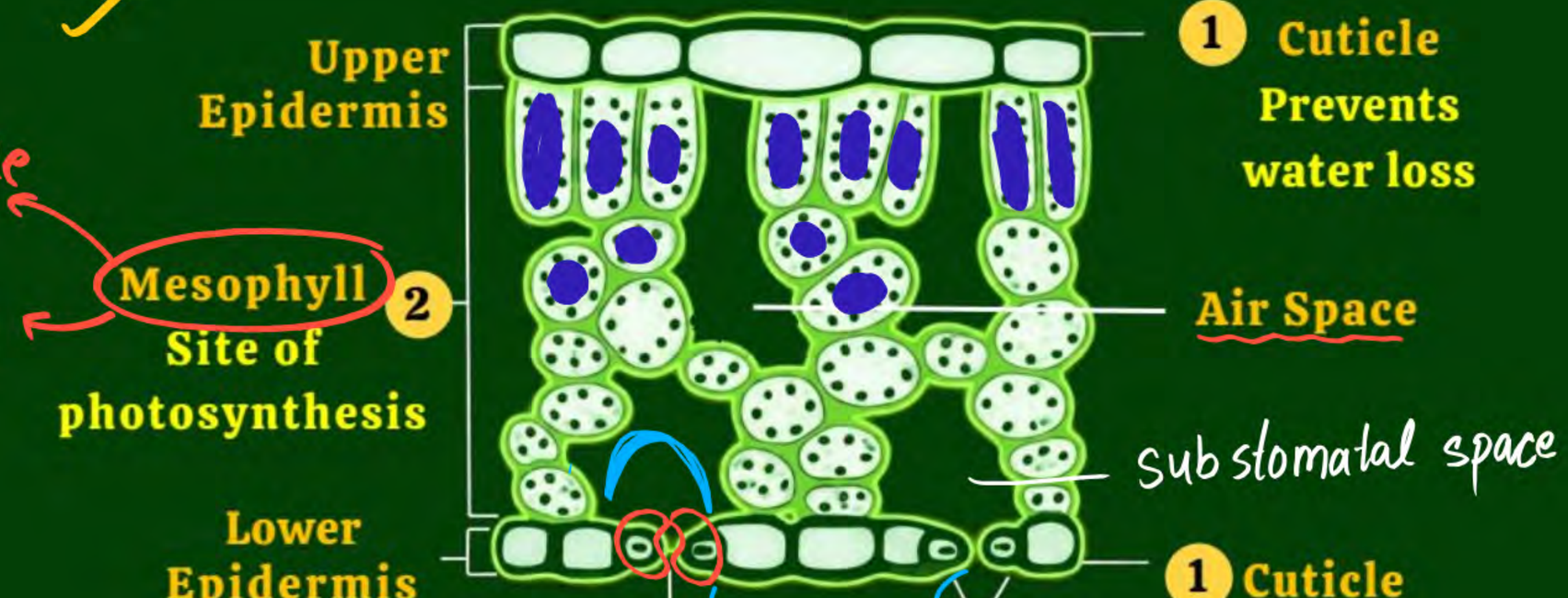
5) Meanwhile, the water from Sub-stomatal space will be lost by transpiration

Original Colour of Cobalt Chloride paper?

A) Blue ✓

B) Pink

MECHANISM OF STOMATAL TRANSPIRATION



QUESTIONS

Label the guidelines A and B

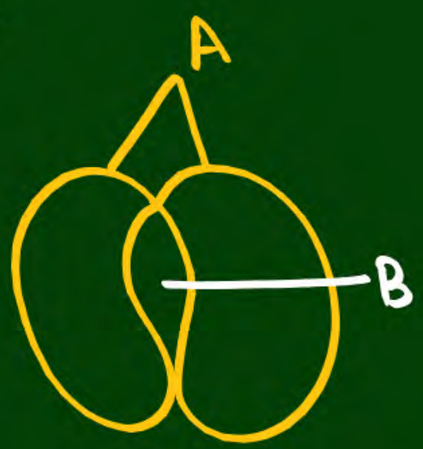
A (A) B (B)

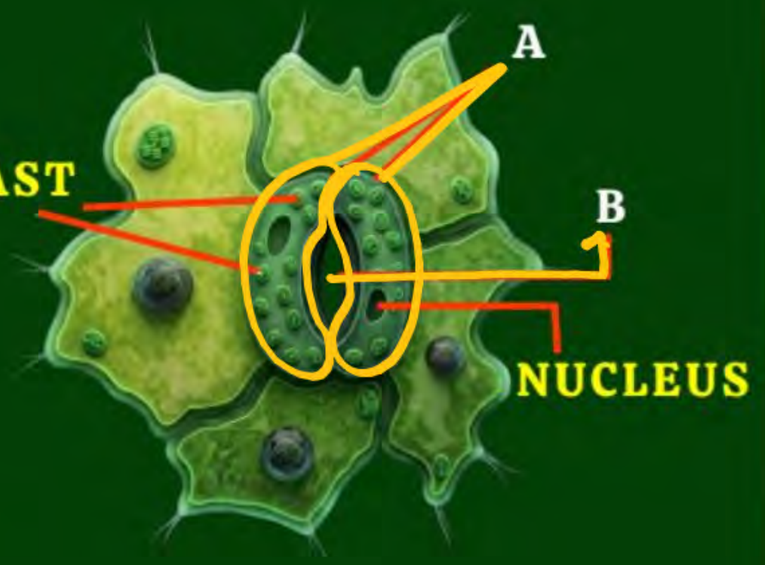
A. Epidermal cells and guard cells

B. Guard cells and stoma

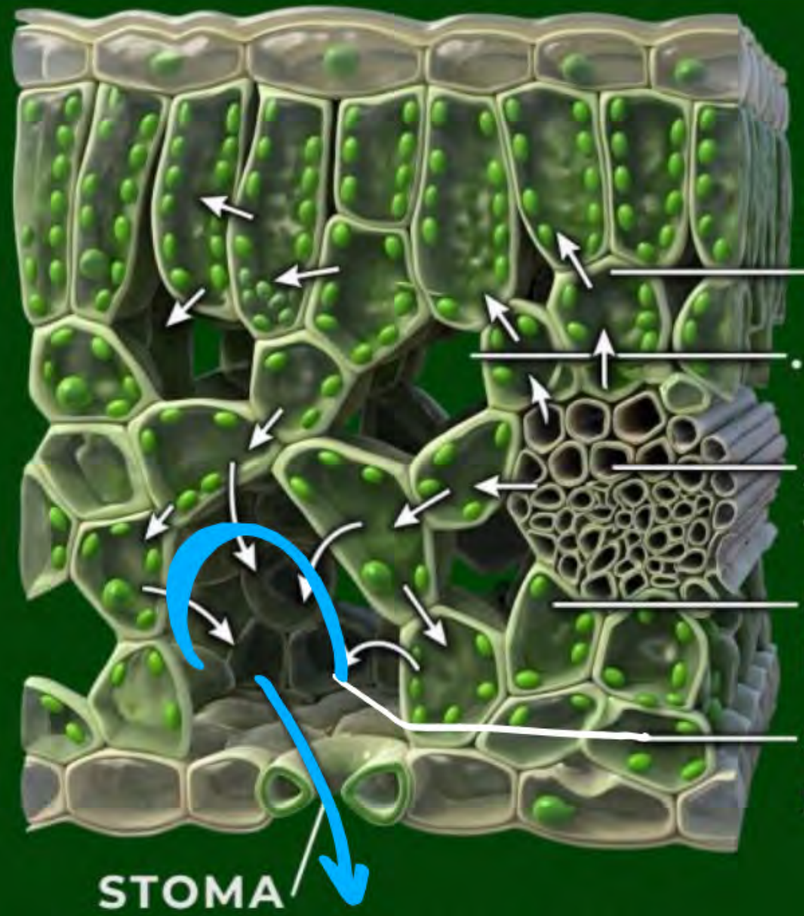
C. Stoma and guard cells

D. Epidermal cells



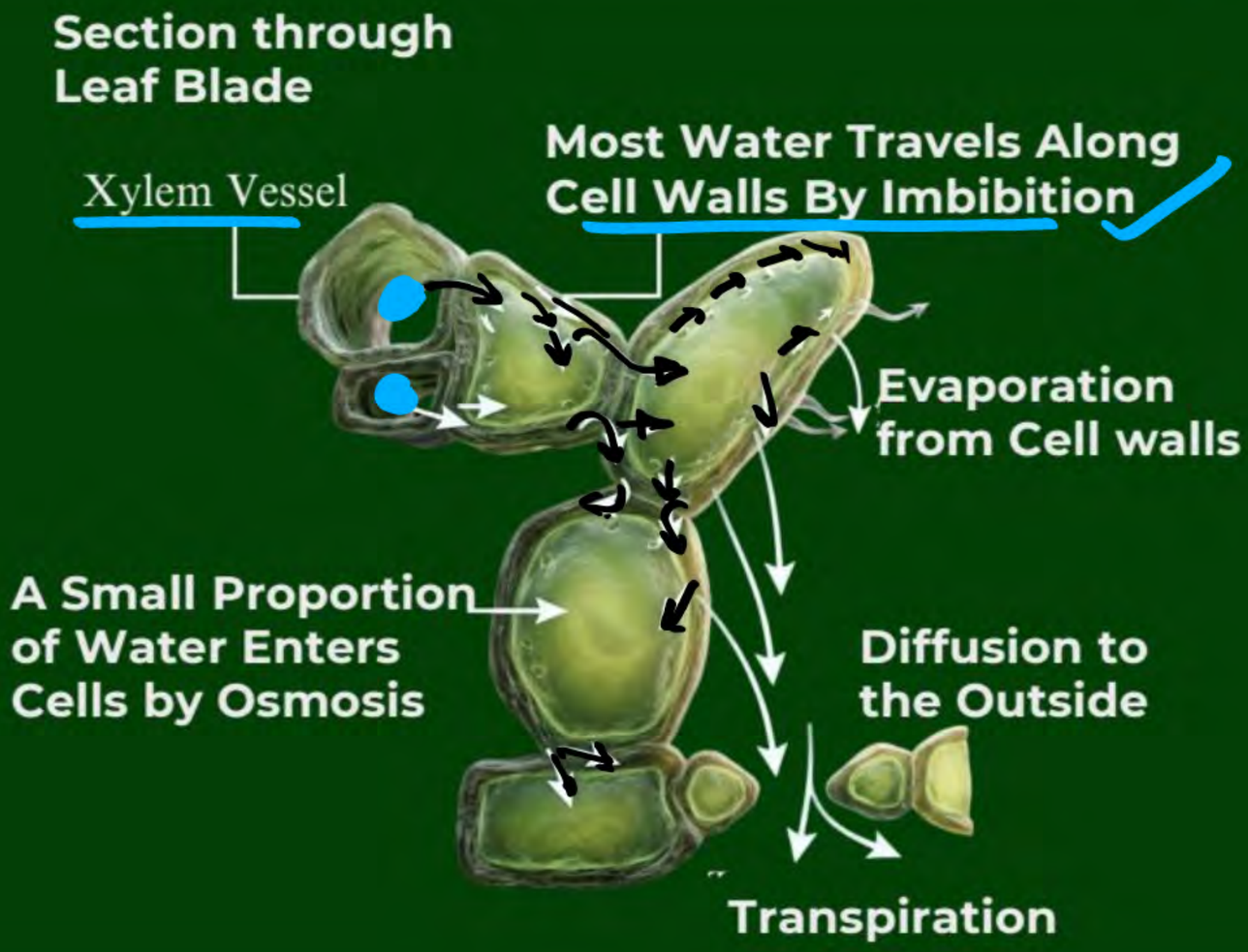


MECHANISM OF STOMATAL TRANSPIRATION



A Section through Leaf Blade

MECHANISM OF STOMATAL TRANSPIRATION



DIFFERENCES BETWEEN STOMATA AND LENTICLES ON THE BASIS OF STRUCTURE, LOCATION AND FUNCTION.

STOMATA	LENTICELS
These are small openings surrounded by two bean-shaped guard cells. They can be opened and closed.	These are special openings which are not surrounded by guard cells. They remain open all the time.
They are mostly found on the abaxial surface of leaves.	These are found on the surface of older woody stems.
These are responsible for diffusion of gases during photosynthesis and respiration. They also help in transpiration.	These are mainly responsible for diffusion of gases during respiration. They also help in transpiration.

Q. The apparatus used to measure rate of transpiration.

A) Manometer

B) Potometer



HOMEWORK

Adaxial → upper

Abaxial → lower

Thank You